In [36]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

data = {
    "Student_ID": [
        101, 102, 103, 104, 105, 106, 107, 108, 109, 110,
        111, 112, 113, 114, 115, 116, 117, 118, 119, 120,
        121, 122, 123, 124, 125,102,105
    ],

    "Name": [
        "  gayathri ", "RAHUL", " priya", "Arun  ", "sNEHA",
        " Kiran ", "DIVYA", " rohit", "ANJALI ", "VAMSI",
        "  meena", "KARTHIK ", " deepa ", "SURESH", "naveen",
        " POOJA ", "ravi  ", " SWATHI", "MANOJ ", " keerthi",
        "  lavanya ", "RAMESH", "bhavya ", " TEJA ", "nithin","RAHUL","sNEHA"
    ],

    "City": [
        " chennai ", "HYDERABAD", " bengaluru", "Delhi ",
        " MUMBAI", "chennai", " hyderabad ", "DELHI",
        " bengaluru ", "Mumbai", " CHENNAI", "hyderabad ",
        "PUNE ", " mumbai", "DELHI ", " bengaluru",
        "CHENNAI ", " pune", "HYDERABAD ", "Mumbai ",
        " delhi", "BENGALURU", " chennai ", "PUNE ", "hyderabad","HYDERABAD","
    ],

    "Age": [
        "21", "22", "twenty", "24", "23", None, "25", "abc",
        "22", "120", "21", "26", "24", "23", "22", "27",
        "25", "24", "21", "23", "22", "28", "24", "26", "25","22","23"
    ],

    "Marks": [
        "85", "90", "78", "88", None, "95", "82", "1000",
        "76", "89", "91", "67", "84", "79", "93", "88",
        "72", "86", "81", "94", "77", "83", "abc", "89", "96","90",None
    ],

    "Fees": [
        "15000", "16000", "15000", "invalid", "17000",
        None, "16000", "18000", "15000", "999999",
        "15500", "16500", "15000", "17000", "16000",
        "18000", "15500", "16500", "15000", "17000",
        "16000", "18000", "15500", "invalid", "16500"
    ],

    "Join_Date": [
        "2025-01-10", "2025-02-15", "2025-03-20",
        "not a date", "2025-04-10", "2025-05-12",
        "2025-06-15", "2025-07-18", None, "2025-09-10",
        "2025-01-18", "2025-02-20", "2025-03-12",
        "2025-04-22", "2025-05-16", "2025-06-11",
        "2025-07-25", "2025-08-14", "2025-09-19",
        "2025-10-10", "2025-01-25", "2025-02-27",
```

```
        "2025-03-18", "2025-04-15", "2025-05-21"
    ],

    "Department": [
        " computer science ", "IT", "COMMERCE", " computer science",
        "IT ", "COMMERCE", " computer science ", "IT",
        "COMMERCE ", " computer science", "IT", "COMMERCE",
        " computer science ", "IT", "COMMERCE", "IT ",
        "computer science", "COMMERCE ", "IT", " computer science",
        "COMMERCE", "IT ", "computer science", "COMMERCE", "IT"
    ]
}

df = pd.DataFrame(data)
print(df)
```

	Student_ID	Name	City	Age	Marks	Fees	Join_Date
0	101	gayathri	chennai	21	85	15000	2025-01-10
1	102	RAHUL	HYDERABAD	22	90	16000	2025-02-15
2	103	priya	bengaluru	twenty	78	15000	2025-03-20
3	104	Arun	Delhi	24	88	invalid	not a date
4	105	sNEHA	MUMBAI	23	None	17000	2025-04-10
5	106	Kiran	chennai	None	95	None	2025-05-12
6	107	DIVYA	hyderabad	25	82	16000	2025-06-15
7	108	rohit	DELHI	abc	1000	18000	2025-07-18
8	109	ANJALI	bengaluru	22	76	15000	None
9	110	VAMSI	Mumbai	120	89	999999	2025-09-10
10	111	meena	CHENNAI	21	91	15500	2025-01-18
11	112	KARTHIK	hyderabad	26	67	16500	2025-02-20
12	113	deepa	PUNE	24	84	15000	2025-03-12
13	114	SURESH	mumbai	23	79	17000	2025-04-22
14	115	naveen	DELHI	22	93	16000	2025-05-16
15	116	POOJA	bengaluru	27	88	18000	2025-06-11
16	117	ravi	CHENNAI	25	72	15500	2025-07-25
17	118	SWATHI	pune	24	86	16500	2025-08-14
18	119	MANOJ	HYDERABAD	21	81	15000	2025-09-19
19	120	keerthi	Mumbai	23	94	17000	2025-10-10
20	121	lavanya	delhi	22	77	16000	2025-01-25
21	122	RAMESH	BENGALURU	28	83	18000	2025-02-27
22	123	bhavya	chennai	24	abc	15500	2025-03-18
23	124	TEJA	PUNE	26	89	invalid	2025-04-15
24	125	nithin	hyderabad	25	96	16500	2025-05-21
25	102	RAHUL	HYDERABAD	22	90	16000	2025-02-15
26	105	sNEHA	MUMBAI	23	None	17000	2025-04-10

	Department
0	computer science
1	IT
2	COMMERCE
3	computer science
4	IT
5	COMMERCE
6	computer science
7	IT
8	COMMERCE
9	computer science
10	IT
11	COMMERCE
12	computer science
13	IT
14	COMMERCE
15	IT
16	computer science
17	COMMERCE
18	IT
19	computer science
20	COMMERCE
21	IT
22	computer science
23	COMMERCE
24	IT

```
25                  IT
26                 IT
```

In [37]:
```
df.head()
```

Out[37]:

	Student_ID	Name	City	Age	Marks	Fees	Join_Date	Department
0	101	gayathri	chennai	21	85	15000	2025-01-10	computer science
1	102	RAHUL	HYDERABAD	22	90	16000	2025-02-15	IT
2	103	priya	bengaluru	twenty	78	15000	2025-03-20	COMMERCE
3	104	Arun	Delhi	24	88	invalid	not a date	computer science
4	105	sNEHA	MUMBAI	23	None	17000	2025-04-10	IT

In [38]:
```
df.shape
```

Out[38]:
```
(27, 8)
```

In [39]:
```
df.columns
```

Out[39]:
```
Index(['Student_ID', 'Name', 'City', 'Age', 'Marks', 'Fees', 'Join_Date',
       'Department'],
      dtype='object')
```

In [40]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 27 entries, 0 to 26
Data columns (total 8 columns):
 #   Column      Non-Null Count  Dtype 
---  ------      --------------  ----- 
 0   Student_ID  27 non-null     int64 
 1   Name        27 non-null     object
 2   City        27 non-null     object
 3   Age         26 non-null     object
 4   Marks       25 non-null     object
 5   Fees        26 non-null     object
 6   Join_Date   26 non-null     object
 7   Department  27 non-null     object
dtypes: int64(1), object(7)
memory usage: 1.8+ KB
```

In [41]:
```
df.describe()
```

Out[41]:

	Student_ID
count	27.000000
mean	112.296296
std	7.523373
min	101.000000
25%	105.500000
50%	112.000000
75%	118.500000
max	125.000000

In [42]:
```
df.isnull()
```

Out[42]:

	Student_ID	Name	City	Age	Marks	Fees	Join_Date	Department
0	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False
4	False	False	False	False	True	False	False	False
5	False	False	False	True	False	True	False	False
6	False	False	False	False	False	False	False	False
7	False	False	False	False	False	False	False	False
8	False	False	False	False	False	False	True	False
9	False	False	False	False	False	False	False	False
10	False	False	False	False	False	False	False	False
11	False	False	False	False	False	False	False	False
12	False	False	False	False	False	False	False	False
13	False	False	False	False	False	False	False	False
14	False	False	False	False	False	False	False	False
15	False	False	False	False	False	False	False	False
16	False	False	False	False	False	False	False	False
17	False	False	False	False	False	False	False	False
18	False	False	False	False	False	False	False	False
19	False	False	False	False	False	False	False	False
20	False	False	False	False	False	False	False	False
21	False	False	False	False	False	False	False	False
22	False	False	False	False	False	False	False	False
23	False	False	False	False	False	False	False	False
24	False	False	False	False	False	False	False	False
25	False	False	False	False	False	False	False	False
26	False	False	False	False	True	False	False	False

In [43]:
```
df.isnull().sum()
```

Out[43]:
```
Student_ID    0
Name          0
City          0
Age           1
Marks         2
Fees          1
Join_Date     1
Department    0
dtype: int64
```

In [44]:
```
df.duplicated()
```

Out[44]:
```
0     False
1     False
2     False
3     False
4     False
5     False
6     False
7     False
8     False
9     False
10    False
11    False
12    False
13    False
14    False
15    False
16    False
17    False
18    False
19    False
20    False
21    False
22    False
23    False
24    False
25     True
26     True
dtype: bool
```

In [45]:
```
df.duplicated().sum()
```

Out[45]:
```
2
```

In [46]:
```
df.dtypes
```

Out[46]:
```
Student_ID     int64
Name          object
City          object
Age           object
Marks         object
Fees          object
Join_Date     object
Department    object
dtype: object
```

In [47]:
```
df["Name"] = df["Name"].str.strip()
df["City"] = df["City"].str.strip()
df["Department"] = df["Department"].str.strip()
```

In [48]:
```
df[["Name", "City", "Department"]]
```

Out[48]:

	Name	City	Department
0	gayathri	chennai	computer science
1	RAHUL	HYDERABAD	IT
2	priya	bengaluru	COMMERCE
3	Arun	Delhi	computer science
4	sNEHA	MUMBAI	IT
5	Kiran	chennai	COMMERCE
6	DIVYA	hyderabad	computer science
7	rohit	DELHI	IT
8	ANJALI	bengaluru	COMMERCE
9	VAMSI	Mumbai	computer science
10	meena	CHENNAI	IT
11	KARTHIK	hyderabad	COMMERCE
12	deepa	PUNE	computer science
13	SURESH	mumbai	IT
14	naveen	DELHI	COMMERCE
15	POOJA	bengaluru	IT
16	ravi	CHENNAI	computer science
17	SWATHI	pune	COMMERCE
18	MANOJ	HYDERABAD	IT
19	keerthi	Mumbai	computer science
20	lavanya	delhi	COMMERCE
21	RAMESH	BENGALURU	IT
22	bhavya	chennai	computer science
23	TEJA	PUNE	COMMERCE
24	nithin	hyderabad	IT
25	RAHUL	HYDERABAD	IT
26	sNEHA	MUMBAI	IT

In [49]:
```
df["Name"] = df["Name"].str.strip().str.title()
df["City"] = df["City"].str.strip().str.title()
df["Department"] = df["Department"].str.strip().str.title()
```

In [50]:
```
df[["Name", "City", "Department"]]
```

Out[50]:

	Name	City	Department
0	Gayathri	Chennai	Computer Science
1	Rahul	Hyderabad	It
2	Priya	Bengaluru	Commerce
3	Arun	Delhi	Computer Science
4	Sneha	Mumbai	It
5	Kiran	Chennai	Commerce
6	Divya	Hyderabad	Computer Science
7	Rohit	Delhi	It
8	Anjali	Bengaluru	Commerce
9	Vamsi	Mumbai	Computer Science
10	Meena	Chennai	It
11	Karthik	Hyderabad	Commerce
12	Deepa	Pune	Computer Science
13	Suresh	Mumbai	It
14	Naveen	Delhi	Commerce
15	Pooja	Bengaluru	It
16	Ravi	Chennai	Computer Science
17	Swathi	Pune	Commerce
18	Manoj	Hyderabad	It
19	Keerthi	Mumbai	Computer Science
20	Lavanya	Delhi	Commerce
21	Ramesh	Bengaluru	It
22	Bhavya	Chennai	Computer Science
23	Teja	Pune	Commerce
24	Nithin	Hyderabad	It
25	Rahul	Hyderabad	It
26	Sneha	Mumbai	It

In [51]:
```
df["Age"] = pd.to_numeric(df["Age"], errors="coerce")
df["Marks"] = pd.to_numeric(df["Marks"], errors="coerce")
df["Fees"] = pd.to_numeric(df["Fees"], errors="coerce")
```

In [52]:
```
df[["Age", "Marks", "Fees"]]
```

Out[52]:

	Age	Marks	Fees
0	21.0	85.0	15000.0
1	22.0	90.0	16000.0
2	NaN	78.0	15000.0
3	24.0	88.0	NaN
4	23.0	NaN	17000.0
5	NaN	95.0	NaN
6	25.0	82.0	16000.0
7	NaN	1000.0	18000.0
8	22.0	76.0	15000.0
9	120.0	89.0	999999.0
10	21.0	91.0	15500.0
11	26.0	67.0	16500.0
12	24.0	84.0	15000.0
13	23.0	79.0	17000.0
14	22.0	93.0	16000.0
15	27.0	88.0	18000.0
16	25.0	72.0	15500.0
17	24.0	86.0	16500.0
18	21.0	81.0	15000.0
19	23.0	94.0	17000.0
20	22.0	77.0	16000.0
21	28.0	83.0	18000.0
22	24.0	NaN	15500.0
23	26.0	89.0	NaN
24	25.0	96.0	16500.0
25	22.0	90.0	16000.0
26	23.0	NaN	17000.0

In [53]:
```
df["Join_Date"] = pd.to_datetime(df["Join_Date"],errors="coerce")
```

In [54]:
```
df[["Join_Date"]]
```

Out[54]:

	Join_Date
0	2025-01-10
1	2025-02-15
2	2025-03-20
3	NaT
4	2025-04-10
5	2025-05-12
6	2025-06-15
7	2025-07-18
8	NaT
9	2025-09-10
10	2025-01-18
11	2025-02-20
12	2025-03-12
13	2025-04-22
14	2025-05-16
15	2025-06-11
16	2025-07-25
17	2025-08-14
18	2025-09-19
19	2025-10-10
20	2025-01-25
21	2025-02-27
22	2025-03-18
23	2025-04-15
24	2025-05-21
25	2025-02-15
26	2025-04-10

In [55]:
```
df.dtypes
```

Out[55]:
```
Student_ID             int64
Name                  object
City                  object
Age                  float64
Marks                float64
Fees                 float64
Join_Date     datetime64[ns]
Department            object
dtype: object
```

In [56]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 27 entries, 0 to 26
Data columns (total 8 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Student_ID  27 non-null     int64
 1   Name        27 non-null     object
 2   City        27 non-null     object
 3   Age         24 non-null     float64
 4   Marks       24 non-null     float64
 5   Fees        24 non-null     float64
 6   Join_Date   25 non-null     datetime64[ns]
 7   Department  27 non-null     object
dtypes: datetime64[ns](1), float64(3), int64(1), object(3)
memory usage: 1.8+ KB
```

In [57]:
```
df.isnull().sum()
```

Out[57]:
```
Student_ID    0
Name          0
City          0
Age           3
Marks         3
Fees          3
Join_Date     2
Department    0
dtype: int64
```

In [58]:
```
df.duplicated().sum()
```

Out[58]:
```
2
```

In [59]:
```
df = df.drop_duplicates()
```

In [60]:
```
df.duplicated().sum()
```

Out[60]:
```
0
```

In [61]:
```
df["Age"] = df["Age"].fillna(df["Age"].mean())
df["Marks"] = df["Marks"].fillna(df["Marks"].mean())
```

In [62]:
```
df[["Age", "Marks"]].isnull().sum()
```

Out[62]:
```
Age      0
Marks    0
dtype: int64
```

In [63]:
```
df["Fees"] = df["Fees"].fillna(15000)
```

In [64]:
```
 df["Fees"].isnull().sum()
```

Out[64]:
```
0
```

In [65]:
```
df["City"] = df["City"].fillna("Unknown")
```

In [66]:
```
df[df["Join_Date"].isnull()]
```

Out[66]:

	Student_ID	Name	City	Age	Marks	Fees	Join_Date	Department
3	104	Arun	Delhi	24.0	88.0	15000.0	NaT	Computer Science
8	109	Anjali	Bengaluru	22.0	76.0	15000.0	NaT	Commerce

In [67]:
```
df = df.dropna(subset=["Join_Date"])
print(df.isnull().sum())
```

```
Student_ID    0
Name          0
City          0
Age           0
Marks         0
Fees          0
Join_Date     0
Department    0
dtype: int64
```

In [68]:
```
Q1 = df["Marks"].quantile(0.25)
Q2 = df["Marks"].quantile(0.50)
Q3 = df["Marks"].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

print("Lower Boundary:", lower_bound)
print("Upper Boundary:", upper_bound)
```

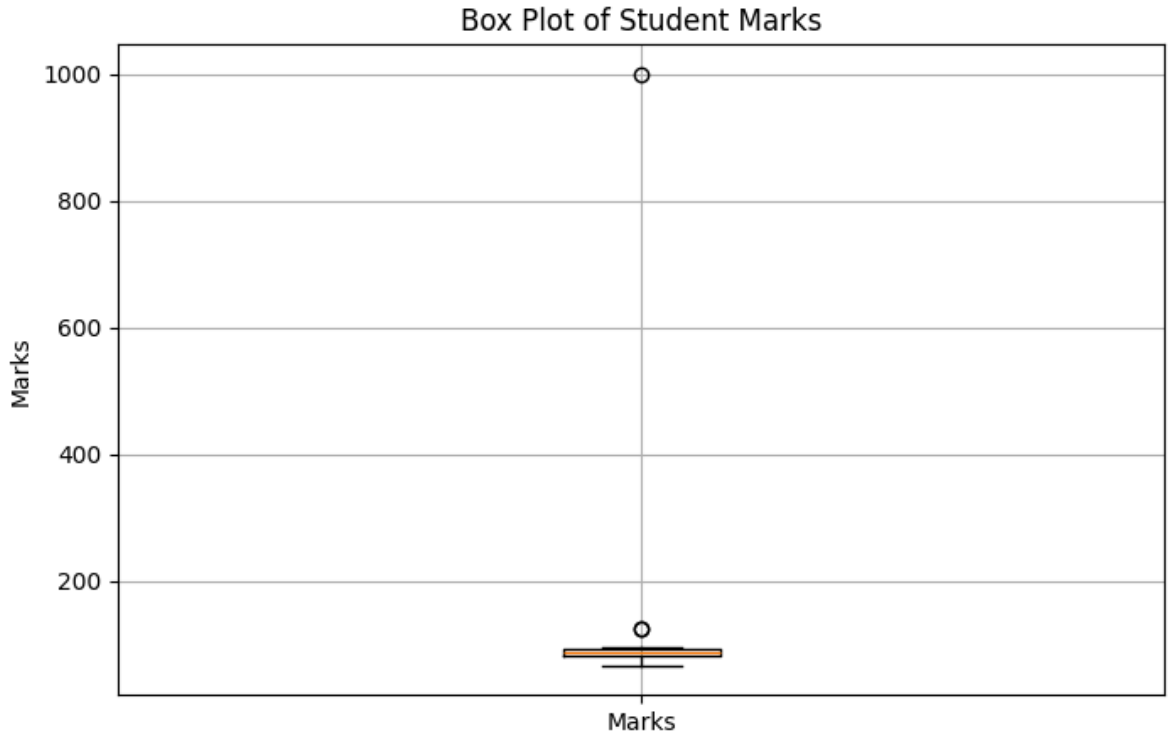
```
Lower Boundary: 63.5
Upper Boundary: 111.5
```

In [69]:
```
outliers = df[(df["Marks"] < lower_bound) |(df["Marks"] > upper_bound)]
print("Outlier rows:")
print(outliers)
```

```
Outlier rows:
    Student_ID    Name     City        Age        Marks     Fees  Join_Date
\
4          105   Sneha   Mumbai  23.000000   124.478261  17000.0 2025-04-10
7          108   Rohit    Delhi  28.090909  1000.000000  18000.0 2025-07-18
22         123  Bhavya  Chennai  24.000000   124.478261  15500.0 2025-03-18

          Department
4                 It
7                 It
22  Computer Science
```

In [70]:
```
plt.figure(figsize=(8, 5))
plt.boxplot(df["Marks"].dropna())
plt.title("Box Plot of Student Marks")
plt.ylabel("Marks")
plt.xticks([1], ["Marks"])
plt.grid(True)
plt.show()
```



In [71]:
```
def assign_grade(marks):
    if marks >= 90:
        return "Excellent"
    elif marks >= 80:
        return "Good"
    elif marks >= 70:
        return "Average"
    else:
        return "Needs Improvement"
```

In [72]:
```
df["Grade"] = df["Marks"].apply(assign_grade)
print(df[["Student_ID", "Name", "Marks", "Grade"]])
```

	Student_ID	Name	Marks	Grade
0	101	Gayathri	85.000000	Good
1	102	Rahul	90.000000	Excellent
2	103	Priya	78.000000	Average
4	105	Sneha	124.478261	Excellent
5	106	Kiran	95.000000	Excellent
6	107	Divya	82.000000	Good
7	108	Rohit	1000.000000	Excellent
9	110	Vamsi	89.000000	Good
10	111	Meena	91.000000	Excellent
11	112	Karthik	67.000000	Needs Improvement
12	113	Deepa	84.000000	Good
13	114	Suresh	79.000000	Average
14	115	Naveen	93.000000	Excellent
15	116	Pooja	88.000000	Good
16	117	Ravi	72.000000	Average
17	118	Swathi	86.000000	Good
18	119	Manoj	81.000000	Good
19	120	Keerthi	94.000000	Excellent
20	121	Lavanya	77.000000	Average
21	122	Ramesh	83.000000	Good
22	123	Bhavya	124.478261	Excellent
23	124	Teja	89.000000	Good
24	125	Nithin	96.000000	Excellent

In [ ]:

In [ ]:

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